

★ UPSTREAM/DOWNSTREAM + SPEEDS ★

Quant | Ch05 Boats + Clocks + Algebra + 2D | Topic 1 of 12

► Meaning in one line

Down = still, stream.

► Key points

Point	Example
$D = b+w$, $U = b-w$.	D15 U9 → b12 w3.
$b = (D+U)/2$, $w = (D-U)/2$.	D20 U12 → b16 w4.

★ IMPORTANT FOR EXAM

Boat-times + clocks + calendar

⚠ TRAP - read twice

Add-for-upstream + 'still = average always' + no-pair/solve

✓ Try yourself: D18 U10. b? w? **Answer:** b14, w4.

★ BOAT TIMES + CLOCK HANDS ★

Quant | Ch05 Boats + Clocks + Algebra + 2D | Topic 2 of 12

► Meaning in one line

Time = dist/speed , upstream/downstream.

► Key points

Point	Example
$t = D/S$ each-way.	60 km, D15 U10: $4+6 = 10$ h.
hour $30^\circ/\text{h}$, min $6^\circ/\text{m}$.	3:00 $\rightarrow 90^\circ$.

★ IMPORTANT FOR EXAM

Angle + coincide + gain/loss

⚠ TRAP - read twice

Speed-add + 'clock = 360/12 only' + minute-miss

✓ **Try yourself:** 72 km, D18 U12 round time? Angle 6:00? **Answer:** $4+6 = 10$ h; 180° .

★ ANGLE + COINCIDE TIMES ★

Quant | Ch05 Boats + Clocks + Algebra + 2D | Topic 3 of 12

► Meaning in one line

Angle = $|30H - 5.5M|$, mod-360.

► Key points

Point	Example
$ 30H - 5.5M $, mod 360.	4:20 $\rightarrow 120 - 110 = 10^\circ$.
coincide every $65\frac{5}{11}$ min ($\sim 11 \times /12h$).	between 7-8 $\approx$ 7:38.

★ IMPORTANT FOR EXAM

Gain/loss + leap + odd-days

⚠ TRAP - read twice

Hour-only + 'coincide hourly' + mod-miss

✓ **Try yourself:** Angle 8:24? Coincide count per 12h? **Answer:** $|240 - 132| = 108^\circ$; 11 times.

★ GAIN/LOSS + LEAP YEARS ★

Quant | Ch05 Boats + Clocks + Algebra + 2D | Topic 4 of 12

► Meaning in one line

Fast/slow , per-day , correct-time.

► Key points

Point	Example
gain/loss per day + correct.	+5 min/day, after 12h shown? actual 11:55... (12h: gain 5/12 min... keep simple: set right daily).
div-4, century div-400.	2000 leap; 1900 not; 2024 leap.

★ IMPORTANT FOR EXAM

Odd-days + day-calc + repeat

⚠ TRAP - read twice

Gain-as-correct + 'century always' + 400-miss

✓ **Try yourself:** Leap? 1800, 2000, 2028? **Answer:** No (century/400); yes; yes.

★ ODD DAYS + DAY CALCULATION ★

Quant | Ch05 Boats + Clocks + Algebra + 2D | Topic 5 of 12

► Meaning in one line

Year/100/400 , month , date , mod-7 , Sunday-Saturday.

► Key points

Point	Example
odd = remainder mod 7.	100 yrs = 5; 400 yrs = 0; ordinary yr = 1; leap yr = 2.
total odd + weekday.	1 Jan 2000 = Saturday (known anchor) → practice with given anchor.

★ IMPORTANT FOR EXAM

Calendar-repeat + linear-equations

⚠ TRAP - read twice

All-months-30 + 'odd = even' + leap-miss + anchor-miss

✓ **Try yourself:** Odd days in 350 days? Ordinary year odd? **Answer:** $350 \bmod 7 = 0$; 1.

★ CALENDAR REPEAT + LINEAR (ONE VAR) ★

Quant | Ch05 Boats + Clocks + Algebra + 2D | Topic 6 of 12

► Meaning in one line

28/40 , leap/ordinary , Jan/Feb-leap.

► Key points

Point	Example
leap +28, ordinary +11/+6 cycle.	2024 leap $\rightarrow$ 2052.
isolate.	$3x+7 = 25 \rightarrow x = 6$; check 25.

★ IMPORTANT FOR EXAM

Two-equations + identities

⚠ TRAP - read twice

Repeat-guess + 'same date' + leap-miss + sign-flip + no-check

✓ **Try yourself:** 2023 repeats? Solve $5x-11 = 24$? **Answer:** 2034 (+11); $x = 7$.

★ TWO LINEAR EQUATIONS + IDENTITIES ★

Quant | Ch05 Boats + Clocks + Algebra + 2D | Topic 7 of 12

► Meaning in one line

Eliminate/substitute.

► Key points

Point	Example
eliminate/substitute.	$x+y = 10$, $x-y = 4 \rightarrow x6, y4$.
$(a+b)^2$, $a^2-b^2 = (a-b)(a+b)$.	$103^2-97^2 = 10000-9 = 9991$.

★ IMPORTANT FOR EXAM

Word-translation + squares

⚠ TRAP - read twice

Add-all + 'solution obvious' + identity-mix

✓ **Try yourself:** $2x+3y = 17$, $x-y = 1$? 98·102? **Answer:** $x4$? $2x+3y=17$, $x = y+1 \rightarrow 5y = 15 \rightarrow y3, x4$; $10000-4 = 9996$.

★ WORD TRANSLATION + SQUARE ★

Quant | Ch05 Boats + Clocks + Algebra + 2D | Topic 8 of 12

► Meaning in one line

English-to-algebra , let , relate , solve.

► Key points

Point	Example
let + relate + solve.	5 more than twice x is 25 $\rightarrow x = 10$.
$A = s^2$, $P = 4s$, $d = s\sqrt{2}$.	P36 $\rightarrow$ s9, A81.

★ IMPORTANT FOR EXAM

Rectangle + triangle

⚠ TRAP - read twice

Numbers-first + 'of = multiply always' + units-mix + no-check

✓ **Try yourself:** 'Thrice less 4 is 20'? Square P28 area? **Answer:** $3x-4 = 20 \rightarrow x=8$; $s^2 \rightarrow 49$.

★ RECTANGLE + TRIANGLE ★

Quant | Ch05 Boats + Clocks + Algebra + 2D | Topic 9 of 12

► Meaning in one line

$L \times W$, $2(L, W)$, diagonal.

► Key points

Point	Example
$A = LW$, $P = 2(L+W)$.	12×5 : A60, P34.
$A = bh/2$; 3-4-5; Heron.	6,8,10 $\rightarrow$ 24.

★ IMPORTANT FOR EXAM

Equilateral + circle

⚠ TRAP - read twice

L+W-area + 'slant = height' + Pythagoras-mix + units

✓ **Try yourself:** 15×8 area/perimeter? Triangle 5,12,13 area? **Answer:** 120, 46; 30.

★ EQUILATERAL + CIRCLE ★

Quant | Ch05 Boats + Clocks + Algebra + 2D | Topic 10 of 12

► Meaning in one line

Root-3/4×side² , height.

► Key points

Point	Example
$A = \frac{\sqrt{3}}{4} a^2$, $h = \frac{\sqrt{3}}{2} a$.	$a6 \rightarrow 9\sqrt{3} \approx 15.6$.
$A = \pi r^2$, $C = 2\pi r$.	$r7 \rightarrow A154, C44$.

★ IMPORTANT FOR EXAM

Semicircle + composite

⚠ TRAP - read twice

Basexheight + 'pi = 3.14 always' + diameter-radius-mix

✓ **Try yourself:** Equilateral 4 area? Circle r14 (22/7)? **Answer:** $4\sqrt{3} \approx 6.9$; A616, C88.

★ SEMICIRCLE + COMPOSITE 2D ★

Quant | Ch05 Boats + Clocks + Algebra + 2D | Topic 11 of 12

► Meaning in one line

Half-circle , half-perimeter , diameter.

► Key points

Point	Example
$A = \pi r^2/2$, $P = \pi r + 2r$.	$r7 \rightarrow A77$, $P36$.
add/subtract + shaded + paths.	square 14 minus circle r7: $196 - 154 = 42$.

★ IMPORTANT FOR EXAM

Conversions + cube

⚠ TRAP - read twice

Full-circle + 'shaded = total' + diameter-miss

✓ **Try yourself:** Semicircle r10 (3.14)? Shaded: 10-square minus r5 circle? **Answer:** A157, P51.4;
 $100 - 78.5 = 21.5$.

★ UNIT CONVERSIONS (LENGTH/AREA) ★

Quant | Ch05 Boats + Clocks + Algebra + 2D | Topic 12 of 12

► Meaning in one line

10/100/1000 , inch/foot/yd/mile , acre/hectare , square.

► Key points

Point	Example
mm-cm-m-km + in/ft/yd/mi.	2.5 km = 2500 m; 12 in = 1 ft.
square units + acre/hectare.	5 acres $\approx$ 20234 m ² (4046.86 $\times$ 5).

★ IMPORTANT FOR EXAM

Cube + cuboid + cylinder

⚠ TRAP - read twice

Linear-for-area + 'acre = fixed-m' + factor-mix

✓ **Try yourself:** 3.2 m in cm? 2 hectares in m²? **Answer:** 320 cm; 20000 m².